

Transient Heat Conduction Time and Temperature Distributions in a Semi-Infinite Solid

Question

We shall examine the temperature in the semi-infinite solid in two ways. In the first way, the temperature is plotted over the range $0 \leq \eta \leq 5$ and $0.01 \leq \tau \leq 3$. In the second way, the temperature is plotted as a function of τ for $0.01 \leq \tau \leq 4$ at $\eta = 0, 1, 2, 3, 4$, and 5 .

Solution

1. Governing Equation

$$\theta(x, t) = \operatorname{erfc} \left(\frac{0.5x}{t} \right) - e^{x+tx^2} \operatorname{erfc} \left(\frac{0.5x}{t} + t \right)$$

Where,

- $\eta = x$ (space similarity variable, non-dimensional)
- $\tau = t$ (time similarity variable, non-dimensional)
- θ = dimensionless temperature

2. Pick Values for Hand Calculation

Let's take a simple case:

$$x = 1.0, t = 0.5$$

3. Calculate First Term

$$A = \frac{0.5x}{t} = \frac{0.5 \cdot 1}{0.5} = 1.0$$

So,

$$\operatorname{erfc}(A) = \operatorname{erfc}(1.0)$$

From tables,

$$\operatorname{erf}(1) = 0.8427$$

$$\operatorname{erfc}(1) = 1 - 0.8427 = 0.1573$$

4. Calculate Second Term

$$B = \frac{0.5x}{t} + t = 1.0 + 0.5 = 1.5$$

So,

$$\operatorname{erf}(1.5).$$

From tables,

$$\operatorname{erf}(1.5) = 0.9661$$

$$\operatorname{erfc}(1.5) = 1 - 0.9661 = 0.0339$$

Now compute the exponential part,

$$e^{x+t^2} = e^{1.0+(0.5)^2} = e^{1.25}$$

$$e^{1.25} \approx 3.4903$$

So, second term,

$$e^{x+t^2} \cdot \operatorname{erfc}(B) = 3.4903 \cdot 0.0339 \approx 0.1183$$

5. Put Together

$$\theta(x, t) = \operatorname{erfc}(1.0) - e^{1.25} \cdot \operatorname{erfc}(1.5)$$

$$\theta(1, 0.5) = 0.1573 - 0.1183 = 0.039$$

Result

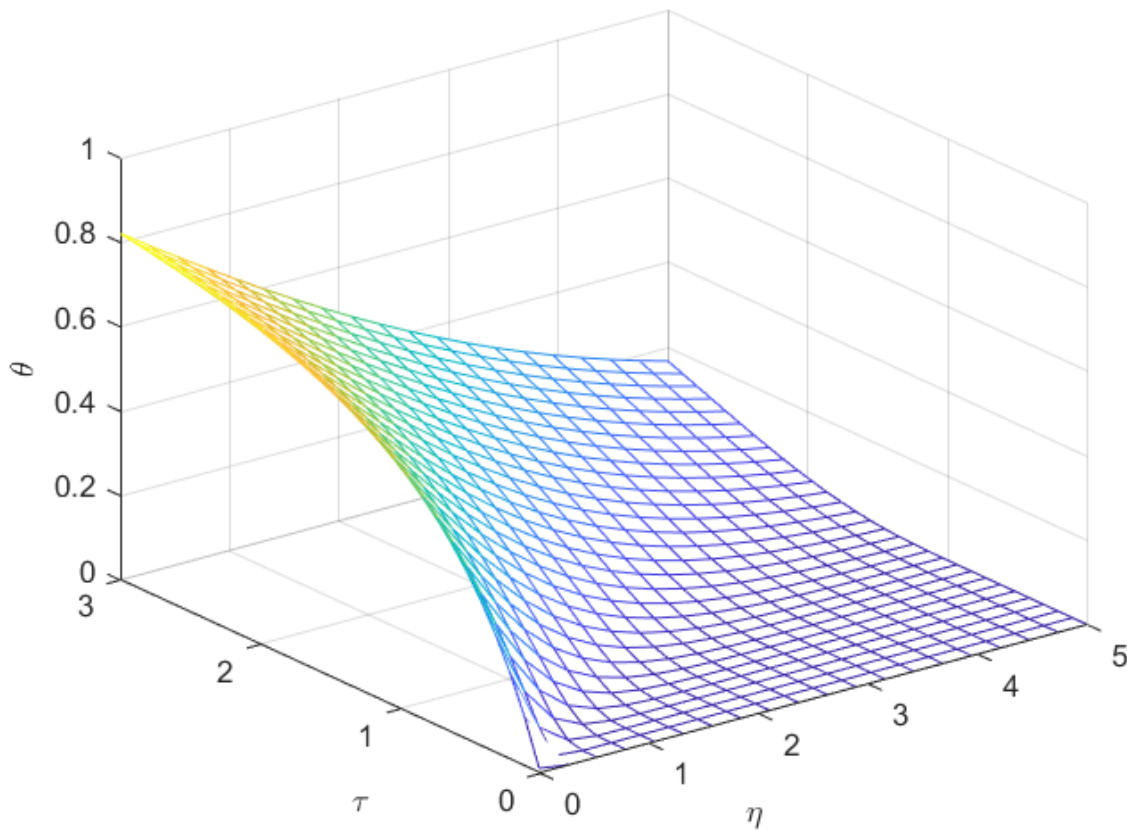


Figure 12.1 Temperature in a semi-infinite solid as a function of nondimensional position η and nondimensional time τ .

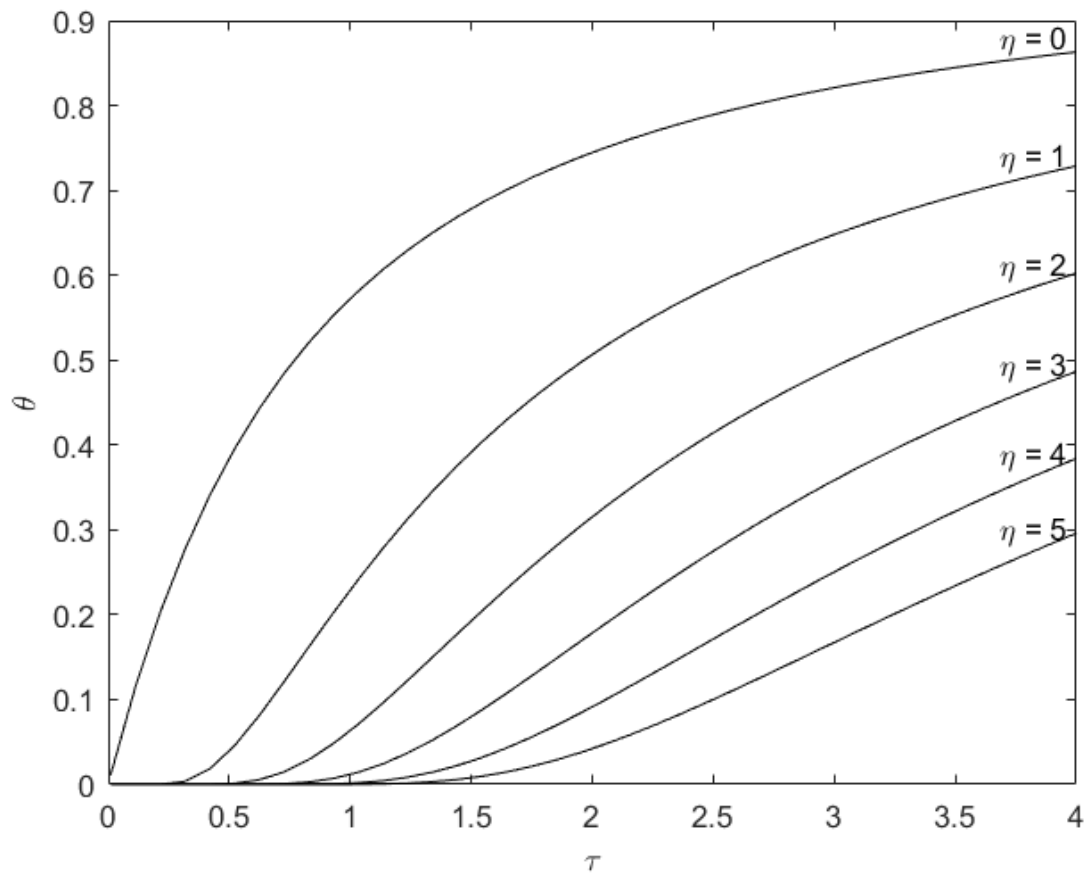


Figure 12.2 Temperature in a semi-infinite solid as a function of nondimensional time τ at several nondimensional locations η .